In [1]:
```
import pandas as pd
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report
import mysql.connector

conn = mysql.connector.connect(
    host='localhost',
    user='root',
    password='Amogh@2003',
    database='air_quality'
)

df2 = pd.read_sql("SELECT * FROM city_hour", conn)
```

```
C:\Users\amogh\AppData\Local\Temp\ipykernel_5252\2131374601.py:14: UserWarning: pandas only supports SQLAlchemy
connectable (engine/connection) or database string URI or sqlite3 DBAPI2 connection. Other DBAPI2 objects are no
t tested. Please consider using SQLAlchemy.
  df2 = pd.read_sql("SELECT * FROM city_hour", conn)
```

In [2]:
```
df2
```

Out[2]:

	ï»¿City	Datetime	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	O3	Benzene	Toluene	Xylene	AQI	AQI_
0	Ahmedabad	01-01-2015			1.00	40.01	36.37		1.00	122.07		0	0	0		
1	Ahmedabad	01-01-2015			0.02	27.75	19.73		0.02	85.90		0	0	0		
2	Ahmedabad	01-01-2015			0.08	19.32	11.08		0.08	52.83		0	0	0		
3	Ahmedabad	01-01-2015			0.30	16.45	9.20		0.30	39.53	153.58	0	0	0		
4	Ahmedabad	01-01-2015			0.12	14.90	7.85		0.12	32.63		0	0	0		
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
53584	Chandigarh	18-03-2020	19.63	71.1	1.73	5.82	4.41	88.06	0.65	7.84	4.04	4	0	1	76	Sat
53585	Chandigarh	18-03-2020	21.78	80.65	2.03	6.17	4.61	84.6	0.64	7.29	2.36	4	0	1	76	Sat
53586	Chandigarh	18-03-2020	21.88	83.69	2.99	6.80	6.04	79.8	0.66	6.07	1.54	5	0	1	77	Sat
53587	Chandigarh	18-03-2020	18.83	79.62	2.49	6.91	5.65	77.32	0.63	4.85	3.92	4	0	1	78	Sat
53588	Chandigarh	18-03-2020	20.06	80.16	2.70	12.18	7.62	106.63	0.61	6.53	5.43	4	0	1	79	Sat

53589 rows × 17 columns

In [3]:
```
print(df2.dtypes)
```

```
ï»¿City       object
Datetime      object
PM2.5         object
PM10          object
NO           float64
NO2          float64
NOx          float64
NH3           object
CO           float64
SO2          float64
O3            object
Benzene        int64
Toluene        int64
Xylene         int64
AQI           object
AQI_Bucket    object
prediction    object
dtype: object
```

In [4]:
```
df2[['AQI', 'PM2.5', 'PM10']] = df2[['AQI', 'PM2.5', 'PM10']].replace('', pd.NA)

df2['AQI'] = pd.to_numeric(df2['AQI'], errors='coerce')
df2['PM2.5'] = pd.to_numeric(df2['PM2.5'], errors='coerce')
df2['PM10'] = pd.to_numeric(df2['PM10'], errors='coerce')
```

```python
df2.dropna(subset=['AQI', 'PM2.5', 'PM10'], inplace=True)

print(df2.dtypes[['AQI', 'PM2.5', 'PM10']])
```

```
AQI       float64
PM2.5     float64
PM10      float64
dtype: object
```

In [14]:

```python
df2
```

Out[14]:

	ï»¿City	Datetime	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	O3	Benzene	Toluene	Xylene	AQI	A
20575	Ahmedabad	15-05-2019	25.91	84.90	13.85	58.41	44.29		13.85	140.10	47.69	17	83	12	255.0	
20576	Ahmedabad	15-05-2019	27.27	98.29	11.30	71.93	47.98		11.30	144.20	71.96	13	80	11	255.0	
20577	Ahmedabad	15-05-2019	27.27	102.68	8.16	66.79	41.89		8.16	124.37	88.85	14	85	16	255.0	
20578	Ahmedabad	15-05-2019	27.30	102.68	7.52	60.64	38.16		7.52	170.01	97.76	18	84	18	255.0	
20579	Ahmedabad	15-05-2019	23.84	121.94	9.06	70.40	44.67		9.06	31.13	94.66	22	92	10	255.0	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
53584	Chandigarh	18-03-2020	19.63	71.10	1.73	5.82	4.41	88.06	0.65	7.84	4.04	4	0	1	76.0	
53585	Chandigarh	18-03-2020	21.78	80.65	2.03	6.17	4.61	84.6	0.64	7.29	2.36	4	0	1	76.0	
53586	Chandigarh	18-03-2020	21.88	83.69	2.99	6.80	6.04	79.8	0.66	6.07	1.54	5	0	1	77.0	
53587	Chandigarh	18-03-2020	18.83	79.62	2.49	6.91	5.65	77.32	0.63	4.85	3.92	4	0	1	78.0	
53588	Chandigarh	18-03-2020	20.06	80.16	2.70	12.18	7.62	106.63	0.61	6.53	5.43	4	0	1	79.0	

29381 rows × 16 columns

In [6]:

```python
X = df2[['NO2', 'PM10', 'CO']]
y = df2['AQI']

from sklearn.preprocessing import LabelEncoder
label_encoder = LabelEncoder()
y = label_encoder.fit_transform(y)

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

rf_model = RandomForestClassifier(n_estimators=100, random_state=42)
rf_model.fit(X_train, y_train)

y_pred = rf_model.predict(X_test)

print(classification_report(y_test, y_pred))

for pred in y_pred:
    query = f"INSERT INTO air_quality.city_hour (prediction) VALUES ({pred})"
    cursor = conn.cursor()
    cursor.execute(query)
    conn.commit()

conn.close()
```

```
C:\Users\amogh\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\metrics\_classification.py:1565
: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use
`zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
C:\Users\amogh\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\metrics\_classification.py:1565
: UndefinedMetricWarning: Recall is ill-defined and being set to 0.0 in labels with no true samples. Use `zero_d
ivision` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
C:\Users\amogh\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\metrics\_classification.py:1565
: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use
`zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
C:\Users\amogh\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\metrics\_classification.py:1565
: UndefinedMetricWarning: Recall is ill-defined and being set to 0.0 in labels with no true samples. Use `zero_d
ivision` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
C:\Users\amogh\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\metrics\_classification.py:1565
: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use
`zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
C:\Users\amogh\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\metrics\_classification.py:1565
: UndefinedMetricWarning: Recall is ill-defined and being set to 0.0 in labels with no true samples. Use `zero_d
ivision` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
              precision    recall  f1-score   support

           1       0.00      0.00      0.00         0
           2       0.00      0.00      0.00         1
           3       0.00      0.00      0.00         2
           4       0.00      0.00      0.00         1
           5       0.07      0.33      0.12         3
           6       0.00      0.00      0.00         2
           7       0.17      0.33      0.22         3
           8       0.00      0.00      0.00         2
           9       0.00      0.00      0.00        12
          10       0.00      0.00      0.00         6
          11       0.25      0.20      0.22        10
          12       0.14      0.17      0.15         6
          13       0.00      0.00      0.00         8
          14       0.11      0.08      0.09        13
          15       0.08      0.12      0.10        16
          16       0.00      0.00      0.00        24
          17       0.05      0.06      0.05        17
          18       0.20      0.14      0.17        14
          19       0.03      0.05      0.04        22
          20       0.07      0.12      0.09         8
          21       0.04      0.04      0.04        28
          22       0.22      0.27      0.24        26
          23       0.09      0.13      0.11        30
          24       0.06      0.07      0.07        27
          25       0.10      0.08      0.09        38
          26       0.00      0.00      0.00        28
          27       0.06      0.06      0.06        31
          28       0.03      0.03      0.03        40
          29       0.07      0.07      0.07        56
          30       0.02      0.04      0.03        24
          31       0.07      0.07      0.07        41
          32       0.09      0.08      0.08        38
          33       0.00      0.00      0.00        39
          34       0.09      0.12      0.10        25
          35       0.06      0.09      0.07        34
          36       0.04      0.05      0.05        40
          37       0.06      0.05      0.06        37
          38       0.02      0.03      0.02        29
          39       0.06      0.06      0.06        50
          40       0.02      0.02      0.02        51
          41       0.05      0.04      0.05        45
          42       0.12      0.17      0.14        41
          43       0.04      0.07      0.05        30
          44       0.03      0.02      0.03        50
          45       0.00      0.00      0.00        27
          46       0.04      0.03      0.03        36
          47       0.00      0.00      0.00        54
          48       0.04      0.03      0.03        32
          49       0.06      0.06      0.06        49
          50       0.00      0.00      0.00        27
          51       0.00      0.00      0.00        49
          52       0.02      0.02      0.02        43
          53       0.05      0.07      0.06        44
          54       0.03      0.04      0.03        28
          55       0.02      0.02      0.02        41
          56       0.00      0.00      0.00        43
          57       0.02      0.02      0.02        56
```

58	0.03	0.03	0.03	33
59	0.03	0.03	0.03	34
60	0.00	0.00	0.00	35
61	0.04	0.04	0.04	49
62	0.00	0.00	0.00	34
63	0.05	0.03	0.03	36
64	0.04	0.04	0.04	26
65	0.00	0.00	0.00	38
66	0.00	0.00	0.00	29
67	0.05	0.05	0.05	41
68	0.00	0.00	0.00	31
69	0.03	0.03	0.03	36
70	0.00	0.00	0.00	29
71	0.00	0.00	0.00	31
72	0.03	0.05	0.04	19
73	0.00	0.00	0.00	49
74	0.06	0.07	0.07	27
75	0.00	0.00	0.00	40
76	0.00	0.00	0.00	38
77	0.05	0.05	0.05	38
78	0.07	0.09	0.07	35
79	0.03	0.02	0.03	41
80	0.05	0.05	0.05	66
81	0.00	0.00	0.00	37
82	0.07	0.06	0.07	47
83	0.09	0.13	0.11	52
84	0.02	0.02	0.02	43
85	0.06	0.08	0.07	38
86	0.02	0.03	0.02	40
87	0.07	0.11	0.09	36
88	0.03	0.03	0.03	34
89	0.02	0.02	0.02	42
90	0.06	0.06	0.06	31
91	0.03	0.03	0.03	38
92	0.07	0.05	0.06	38
93	0.05	0.04	0.04	27
94	0.00	0.00	0.00	26
95	0.00	0.00	0.00	23
96	0.00	0.00	0.00	21
97	0.00	0.00	0.00	25
98	0.03	0.03	0.03	33
99	0.00	0.00	0.00	30
100	0.04	0.03	0.03	33
101	0.00	0.00	0.00	25
102	0.00	0.00	0.00	24
103	0.00	0.00	0.00	33
104	0.00	0.00	0.00	31
105	0.00	0.00	0.00	29
106	0.03	0.03	0.03	29
107	0.06	0.05	0.06	40
108	0.03	0.03	0.03	31
109	0.09	0.06	0.07	31
110	0.00	0.00	0.00	23
111	0.00	0.00	0.00	21
112	0.00	0.00	0.00	16
113	0.06	0.08	0.07	12
114	0.00	0.00	0.00	19
115	0.00	0.00	0.00	20
116	0.04	0.03	0.04	30
117	0.12	0.09	0.11	55
118	0.03	0.03	0.03	40
119	0.09	0.06	0.07	51
120	0.05	0.05	0.05	37
121	0.00	0.00	0.00	23
122	0.08	0.07	0.08	27
123	0.00	0.00	0.00	20
124	0.11	0.04	0.06	23
125	0.00	0.00	0.00	27
126	0.00	0.00	0.00	11
127	0.00	0.00	0.00	9
128	0.00	0.00	0.00	12
129	0.11	0.09	0.10	23
130	0.00	0.00	0.00	14
131	0.14	0.08	0.10	13
132	0.00	0.00	0.00	8
133	0.00	0.00	0.00	18
134	0.12	0.15	0.13	13
135	0.00	0.00	0.00	18
136	0.18	0.11	0.13	19
137	0.00	0.00	0.00	12
138	0.00	0.00	0.00	18
139	0.00	0.00	0.00	18
140	0.00	0.00	0.00	18

141	0.00	0.00	0.00	17
142	0.00	0.00	0.00	19
143	0.06	0.10	0.07	10
144	0.07	0.05	0.06	19
145	0.05	0.06	0.06	16
146	0.00	0.00	0.00	9
147	0.00	0.00	0.00	11
148	0.00	0.00	0.00	14
149	0.08	0.09	0.08	11
150	0.00	0.00	0.00	14
151	0.00	0.00	0.00	9
152	0.00	0.00	0.00	10
153	0.00	0.00	0.00	4
154	0.14	0.08	0.10	13
155	0.00	0.00	0.00	5
156	0.00	0.00	0.00	11
157	0.00	0.00	0.00	5
158	0.00	0.00	0.00	15
159	0.00	0.00	0.00	8
160	0.00	0.00	0.00	19
161	0.00	0.00	0.00	13
162	0.00	0.00	0.00	8
163	0.00	0.00	0.00	5
164	0.00	0.00	0.00	8
165	0.00	0.00	0.00	10
166	0.00	0.00	0.00	7
167	0.00	0.00	0.00	8
168	0.00	0.00	0.00	6
169	0.00	0.00	0.00	3
170	0.00	0.00	0.00	11
171	0.00	0.00	0.00	14
172	0.00	0.00	0.00	8
173	0.00	0.00	0.00	11
174	0.00	0.00	0.00	9
175	0.00	0.00	0.00	9
176	0.00	0.00	0.00	8
177	0.00	0.00	0.00	7
178	0.00	0.00	0.00	9
179	0.00	0.00	0.00	7
180	0.00	0.00	0.00	11
181	0.00	0.00	0.00	7
182	0.00	0.00	0.00	3
183	0.00	0.00	0.00	11
184	0.00	0.00	0.00	8
185	0.00	0.00	0.00	6
186	0.00	0.00	0.00	5
187	0.00	0.00	0.00	5
188	0.00	0.00	0.00	5
189	0.00	0.00	0.00	3
190	0.06	0.17	0.08	6
191	0.00	0.00	0.00	4
192	0.11	0.09	0.10	11
193	0.00	0.00	0.00	9
194	0.00	0.00	0.00	5
195	0.08	0.10	0.09	10
196	0.00	0.00	0.00	9
197	0.00	0.00	0.00	7
198	0.00	0.00	0.00	6
199	0.00	0.00	0.00	7
200	0.00	0.00	0.00	7
201	0.00	0.00	0.00	6
202	0.00	0.00	0.00	6
203	0.00	0.00	0.00	4
204	0.00	0.00	0.00	11
205	0.25	0.12	0.17	8
206	0.00	0.00	0.00	7
207	0.17	0.25	0.20	4
208	0.00	0.00	0.00	2
209	0.00	0.00	0.00	9
210	0.00	0.00	0.00	3
211	0.00	0.00	0.00	6
212	0.00	0.00	0.00	4
213	0.00	0.00	0.00	5
214	0.00	0.00	0.00	7
215	0.00	0.00	0.00	3
216	0.00	0.00	0.00	6
217	0.00	0.00	0.00	2
218	0.00	0.00	0.00	1
219	0.00	0.00	0.00	7
220	0.11	0.33	0.17	3
221	0.00	0.00	0.00	4
222	0.33	0.40	0.36	5
223	0.00	0.00	0.00	8

224	0.00	0.00	0.00	6
225	0.00	0.00	0.00	3
226	0.00	0.00	0.00	6
227	0.00	0.00	0.00	7
228	0.00	0.00	0.00	6
229	0.00	0.00	0.00	6
230	0.00	0.00	0.00	6
231	0.00	0.00	0.00	8
232	0.00	0.00	0.00	12
233	0.00	0.00	0.00	9
234	0.00	0.00	0.00	4
235	0.00	0.00	0.00	8
236	0.00	0.00	0.00	4
237	0.00	0.00	0.00	1
238	0.00	0.00	0.00	11
239	0.00	0.00	0.00	9
240	0.00	0.00	0.00	10
241	0.00	0.00	0.00	5
242	0.00	0.00	0.00	4
243	0.00	0.00	0.00	2
244	0.00	0.00	0.00	5
245	0.00	0.00	0.00	3
246	0.00	0.00	0.00	5
247	0.00	0.00	0.00	6
248	0.00	0.00	0.00	6
249	0.17	0.12	0.14	8
250	0.00	0.00	0.00	3
251	0.00	0.00	0.00	4
252	0.00	0.00	0.00	4
253	0.00	0.00	0.00	1
254	0.00	0.00	0.00	3
255	0.00	0.00	0.00	5
256	0.00	0.00	0.00	4
257	0.20	0.12	0.15	8
258	0.00	0.00	0.00	3
259	0.25	0.12	0.17	8
260	0.50	0.20	0.29	5
261	0.00	0.00	0.00	3
262	0.20	1.00	0.33	1
263	0.00	0.00	0.00	4
264	0.00	0.00	0.00	4
265	0.00	0.00	0.00	2
266	0.00	0.00	0.00	6
267	0.00	0.00	0.00	3
268	0.00	0.00	0.00	1
269	0.00	0.00	0.00	1
270	0.00	0.00	0.00	1
271	0.00	0.00	0.00	2
272	0.00	0.00	0.00	2
273	0.00	0.00	0.00	4
274	0.00	0.00	0.00	2
275	0.00	0.00	0.00	0
276	0.00	0.00	0.00	4
277	0.00	0.00	0.00	6
278	0.50	0.17	0.25	6
279	0.00	0.00	0.00	4
280	0.00	0.00	0.00	15
281	0.00	0.00	0.00	6
282	0.15	0.14	0.15	14
283	0.00	0.00	0.00	9
284	0.08	0.11	0.10	9
285	0.07	0.05	0.06	20
286	0.00	0.00	0.00	10
287	0.00	0.00	0.00	11
288	0.00	0.00	0.00	6
289	0.00	0.00	0.00	9
290	0.00	0.00	0.00	10
291	0.00	0.00	0.00	10
292	0.09	0.12	0.11	8
293	0.11	0.17	0.13	12
294	0.00	0.00	0.00	12
295	0.00	0.00	0.00	1
296	0.00	0.00	0.00	12
297	0.00	0.00	0.00	11
298	0.06	0.07	0.06	14
299	0.17	0.25	0.20	8
300	0.00	0.00	0.00	13
301	0.00	0.00	0.00	16
302	0.00	0.00	0.00	11
303	0.00	0.00	0.00	10
304	0.00	0.00	0.00	11
305	0.00	0.00	0.00	5
306	0.00	0.00	0.00	5

307	0.25	0.20	0.22	5
308	0.00	0.00	0.00	7
309	0.17	0.20	0.18	5
310	0.00	0.00	0.00	6
311	0.00	0.00	0.00	0
312	0.00	0.00	0.00	1
313	0.00	0.00	0.00	3
314	0.00	0.00	0.00	3
315	0.00	0.00	0.00	3
316	0.00	0.00	0.00	3
317	0.00	0.00	0.00	4
318	0.00	0.00	0.00	2
319	0.00	0.00	0.00	3
320	0.00	0.00	0.00	3
321	0.00	0.00	0.00	5
322	0.00	0.00	0.00	3
323	0.14	0.12	0.13	8
324	0.00	0.00	0.00	6
325	0.00	0.00	0.00	4
326	0.00	0.00	0.00	2
327	0.00	0.00	0.00	4
328	0.00	0.00	0.00	0
329	0.00	0.00	0.00	2
330	0.00	0.00	0.00	0
331	0.00	0.00	0.00	2
332	0.00	0.00	0.00	4
333	0.00	0.00	0.00	2
334	0.00	0.00	0.00	1
335	0.00	0.00	0.00	2
336	0.00	0.00	0.00	0
337	0.00	0.00	0.00	6
338	0.00	0.00	0.00	7
339	0.00	0.00	0.00	3
340	0.00	0.00	0.00	4
341	0.00	0.00	0.00	2
342	0.00	0.00	0.00	2
343	0.00	0.00	0.00	2
344	0.00	0.00	0.00	4
345	0.00	0.00	0.00	5
346	0.00	0.00	0.00	3
347	0.00	0.00	0.00	2
348	0.00	0.00	0.00	2
349	0.00	0.00	0.00	3
350	0.00	0.00	0.00	2
351	0.00	0.00	0.00	1
352	0.00	0.00	0.00	3
353	0.00	0.00	0.00	3
354	0.00	0.00	0.00	1
355	0.00	0.00	0.00	3
356	0.00	0.00	0.00	1
357	0.00	0.00	0.00	2
358	0.08	0.17	0.11	6
359	0.00	0.00	0.00	1
360	0.00	0.00	0.00	3
361	0.50	0.17	0.25	6
363	0.00	0.00	0.00	1
364	0.00	0.00	0.00	0
365	0.00	0.00	0.00	3
366	0.00	0.00	0.00	1
367	0.00	0.00	0.00	0
368	0.00	0.00	0.00	1
369	0.00	0.00	0.00	4
370	0.00	0.00	0.00	3
371	0.00	0.00	0.00	2
372	0.00	0.00	0.00	3
373	0.00	0.00	0.00	2
374	0.00	0.00	0.00	1
375	0.00	0.00	0.00	2
376	0.00	0.00	0.00	2
377	0.00	0.00	0.00	8
378	0.00	0.00	0.00	1
379	0.00	0.00	0.00	2
380	0.00	0.00	0.00	1
381	0.00	0.00	0.00	1
382	0.00	0.00	0.00	3
383	0.00	0.00	0.00	8
384	0.00	0.00	0.00	6
386	0.00	0.00	0.00	1
387	0.00	0.00	0.00	2
388	0.00	0.00	0.00	1
390	0.00	0.00	0.00	0
391	0.00	0.00	0.00	1
392	0.00	0.00	0.00	3

393	0.00	0.00	0.00	2
394	0.00	0.00	0.00	5
395	0.00	0.00	0.00	1
396	0.00	0.00	0.00	0
397	0.00	0.00	0.00	2
398	0.00	0.00	0.00	1
399	0.00	0.00	0.00	0
400	0.33	0.33	0.33	3
401	0.00	0.00	0.00	6
402	0.00	0.00	0.00	3
403	0.00	0.00	0.00	6
404	0.00	0.00	0.00	1
405	0.00	0.00	0.00	2
407	0.00	0.00	0.00	6
408	0.00	0.00	0.00	2
409	0.25	0.25	0.25	4
411	0.00	0.00	0.00	4
412	0.00	0.00	0.00	0
413	0.00	0.00	0.00	1
414	0.33	1.00	0.50	1
415	0.00	0.00	0.00	1
416	0.00	0.00	0.00	1
418	0.00	0.00	0.00	2
419	0.00	0.00	0.00	1
420	0.00	0.00	0.00	2
421	0.00	0.00	0.00	0
422	0.00	0.00	0.00	3
423	0.00	0.00	0.00	1
424	0.00	0.00	0.00	0
425	0.00	0.00	0.00	1
426	0.00	0.00	0.00	1
428	0.00	0.00	0.00	3
429	0.00	0.00	0.00	1
431	0.00	0.00	0.00	1
432	0.00	0.00	0.00	2
433	0.00	0.00	0.00	2
434	0.00	0.00	0.00	0
435	0.00	0.00	0.00	1
436	0.00	0.00	0.00	3
437	0.00	0.00	0.00	1
438	0.00	0.00	0.00	2
439	0.00	0.00	0.00	0
440	0.00	0.00	0.00	0
442	0.50	0.67	0.57	3
443	0.00	0.00	0.00	3
444	0.00	0.00	0.00	3
445	0.00	0.00	0.00	0
447	0.00	0.00	0.00	2
448	0.00	0.00	0.00	3
449	0.00	0.00	0.00	0
450	0.00	0.00	0.00	1
451	0.00	0.00	0.00	2
452	0.00	0.00	0.00	4
454	0.00	0.00	0.00	2
455	0.00	0.00	0.00	1
456	0.00	0.00	0.00	2
457	0.00	0.00	0.00	4
459	0.00	0.00	0.00	6
460	0.00	0.00	0.00	3
461	0.14	0.25	0.18	4
463	0.00	0.00	0.00	0
464	0.00	0.00	0.00	1
465	0.20	0.50	0.29	2
466	0.50	1.00	0.67	2
467	0.50	0.33	0.40	3
468	0.00	0.00	0.00	1
469	0.00	0.00	0.00	4
470	0.00	0.00	0.00	1
471	0.00	0.00	0.00	4
472	0.00	0.00	0.00	5
474	0.00	0.00	0.00	3
475	0.00	0.00	0.00	1
476	0.00	0.00	0.00	0
477	0.00	0.00	0.00	2
478	0.00	0.00	0.00	2
479	0.00	0.00	0.00	2
480	0.00	0.00	0.00	3
483	0.00	0.00	0.00	2
484	0.00	0.00	0.00	0
485	0.00	0.00	0.00	1
486	0.00	0.00	0.00	1
487	0.00	0.00	0.00	0
488	0.00	0.00	0.00	2

489	0.00	0.00	0.00	0
490	0.00	0.00	0.00	1
491	0.00	0.00	0.00	1
492	0.00	0.00	0.00	3
493	0.00	0.00	0.00	2
494	0.00	0.00	0.00	1
495	0.00	0.00	0.00	1
496	0.00	0.00	0.00	3
497	0.00	0.00	0.00	5
498	0.00	0.00	0.00	1
499	0.00	0.00	0.00	3
500	0.00	0.00	0.00	2
501	0.00	0.00	0.00	2
502	0.25	0.33	0.29	3
503	0.00	0.00	0.00	0
504	0.00	0.00	0.00	0
505	0.00	0.00	0.00	1
507	0.00	0.00	0.00	5
508	0.50	0.50	0.50	2
509	0.67	1.00	0.80	2
510	0.00	0.00	0.00	3
512	0.00	0.00	0.00	5
513	0.50	0.33	0.40	3
514	0.00	0.00	0.00	1
515	0.00	0.00	0.00	1
516	0.00	0.00	0.00	2
517	0.00	0.00	0.00	0
518	0.00	0.00	0.00	1
519	0.00	0.00	0.00	1
521	0.25	0.33	0.29	3
523	0.00	0.00	0.00	0
524	0.00	0.00	0.00	2
525	0.00	0.00	0.00	4
526	0.00	0.00	0.00	4
527	0.00	0.00	0.00	1
528	0.00	0.00	0.00	2
529	0.00	0.00	0.00	0
530	0.00	0.00	0.00	1
532	0.00	0.00	0.00	1
535	0.00	0.00	0.00	3
536	0.00	0.00	0.00	1
537	0.00	0.00	0.00	3
538	0.00	0.00	0.00	1
539	0.00	0.00	0.00	1
540	0.00	0.00	0.00	1
541	0.00	0.00	0.00	1
543	0.00	0.00	0.00	1
544	0.00	0.00	0.00	1
545	0.00	0.00	0.00	0
546	0.00	0.00	0.00	0
548	0.00	0.00	0.00	1
549	0.00	0.00	0.00	0
550	0.00	0.00	0.00	3
551	0.00	0.00	0.00	3
552	0.00	0.00	0.00	0
553	0.00	0.00	0.00	2
554	0.00	0.00	0.00	1
555	0.00	0.00	0.00	4
558	0.00	0.00	0.00	1
559	0.00	0.00	0.00	1
561	0.50	0.33	0.40	3
562	0.00	0.00	0.00	0
564	0.00	0.00	0.00	0
565	0.00	0.00	0.00	0
566	0.00	0.00	0.00	2
567	0.00	0.00	0.00	2
569	0.00	0.00	0.00	5
572	0.00	0.00	0.00	3
573	0.00	0.00	0.00	2
576	0.00	0.00	0.00	1
577	0.50	0.12	0.20	8
578	0.00	0.00	0.00	1
579	0.00	0.00	0.00	2
580	0.00	0.00	0.00	1
581	0.00	0.00	0.00	0
583	0.00	0.00	0.00	1
585	0.00	0.00	0.00	3
587	0.00	0.00	0.00	1
588	0.00	0.00	0.00	1
589	0.00	0.00	0.00	1
592	0.00	0.00	0.00	1
597	0.00	0.00	0.00	1
598	0.00	0.00	0.00	0

```
         599       0.00      0.00      0.00         1
         601       0.00      0.00      0.00         0
         602       0.33      0.50      0.40         2
         603       0.00      0.00      0.00         1
         604       1.00      0.50      0.67         2
         605       0.00      0.00      0.00         1
         607       0.00      0.00      0.00         0
         608       0.00      0.00      0.00         1
         609       0.00      0.00      0.00         1
         610       0.00      0.00      0.00         0
         615       0.00      0.00      0.00         1
         618       0.00      0.00      0.00         2
         621       0.00      0.00      0.00         3
         622       0.00      0.00      0.00         1
         625       0.00      0.00      0.00         1
         626       0.00      0.00      0.00         0
         627       0.00      0.00      0.00         1
         630       0.00      0.00      0.00         2
         632       0.00      0.00      0.00         1
         635       0.00      0.00      0.00         2
         636       0.00      0.00      0.00         1
         637       0.00      0.00      0.00         2
         638       0.00      0.00      0.00         0
         639       0.00      0.00      0.00         2
         640       0.00      0.00      0.00         0
         641       0.00      0.00      0.00         1
         642       0.00      0.00      0.00         1
         643       0.00      0.00      0.00         1
         644       0.00      0.00      0.00         2
         646       0.00      0.00      0.00         1
         647       0.50      1.00      0.67         1
         648       0.00      0.00      0.00         1
         649       0.00      0.00      0.00         2
         650       0.00      0.00      0.00         0
         651       0.00      0.00      0.00         0
         652       0.00      0.00      0.00         0
         653       0.00      0.00      0.00         2
         656       1.00      0.17      0.29         6
         657       0.00      0.00      0.00         2
         660       0.00      0.00      0.00         0
         661       0.00      0.00      0.00         0
         662       0.50      1.00      0.67         1
         667       0.00      0.00      0.00         1
         668       0.00      0.00      0.00         1
         669       0.00      0.00      0.00         1
         670       0.00      0.00      0.00         0
         672       0.00      0.00      0.00         1
         674       0.00      0.00      0.00         3
         675       0.00      0.00      0.00         4
         676       0.00      0.00      0.00         0
         677       0.00      0.00      0.00         1
         679       0.00      0.00      0.00         4
         680       0.00      0.00      0.00         1
         683       0.00      0.00      0.00         0
         685       0.00      0.00      0.00         0
         687       0.00      0.00      0.00         1
         689       0.00      0.00      0.00         1
         693       0.00      0.00      0.00         0
         695       0.00      0.00      0.00         2
         696       0.00      0.00      0.00         1
         697       0.00      0.00      0.00         1
         701       0.00      0.00      0.00         0
         703       0.00      0.00      0.00         1
         704       1.00      0.50      0.67         2
         707       0.00      0.00      0.00         0
         708       0.00      0.00      0.00         1
         709       0.33      0.50      0.40         2
         711       0.00      0.00      0.00         1
         712       0.00      0.00      0.00         1
         713       0.00      0.00      0.00         0
         714       0.00      0.00      0.00         2
         716       0.00      0.00      0.00         2
         720       0.00      0.00      0.00         0

    accuracy                           0.04      5877
   macro avg       0.03      0.04      0.03      5877
weighted avg       0.04      0.04      0.04      5877
```

In [8]:
```
import mysql.connector
conn = mysql.connector.connect(
    host='localhost',
    user='root',
```

```
    password='Amogh@2003',
    database='air_quality'
)

cursor = conn.cursor()

create_table_query = """
CREATE TABLE IF NOT EXISTS predicted_aqi (
    id INT AUTO_INCREMENT PRIMARY KEY,
    prediction INT
);
"""

cursor.execute(create_table_query)
conn.commit()

print("Table 'predicted_aqi' created successfully.")

cursor.close()
conn.close()
```

```
Table 'predicted_aqi' created successfully.
```

In [10]:

```
import mysql.connector

conn = mysql.connector.connect(
    host='localhost',
    user='root',
    password='Amogh@2003',
    database='air_quality'
)

cursor = conn.cursor()

cursor.execute("DROP TABLE IF EXISTS predicted_aqi")

cursor.execute("""
CREATE TABLE predicted_aqi (
    id INT AUTO_INCREMENT PRIMARY KEY,
    prediction INT,
    model_used VARCHAR(50),
    predicted_at DATETIME DEFAULT CURRENT_TIMESTAMP
)
""")

conn.commit()
print("Upgraded table 'predicted_aqi' created successfully.")

cursor.close()
conn.close()
```

```
Upgraded table 'predicted_aqi' created successfully.
```

In [11]:

```
import mysql.connector

conn = mysql.connector.connect(
     host='localhost',
    user='root',
    password='Amogh@2003',
    database='air_quality'
)

cursor = conn.cursor()
insert_query = "INSERT INTO predicted_aqi (prediction, model_used) VALUES (%s, %s)"

for pred in y_pred:
    cursor.execute(insert_query, (int(pred), 'RandomForest'))

conn.commit()
cursor.close()
conn.close()

print("Predictions inserted successfully with model info.")
```

```
Predictions inserted successfully with model info.
```

In [14]:

```
conn.close()
```

In [15]:

```
import pandas as pd
import mysql.connector

conn = mysql.connector.connect(
    host='localhost',
    user='root',
```

```python
    password='Amogh@2003',
    database='air_quality'
)

df2 = pd.read_sql("SELECT * FROM city_hour", conn)
```

```
C:\Users\amogh\AppData\Local\Temp\ipykernel_5252\1724708832.py:11: UserWarning: pandas only supports SQLAlchemy connectable (engine/connection) or database string URI or sqlite3 DBAPI2 connection. Other DBAPI2 objects are no
t tested. Please consider using SQLAlchemy.
  df2 = pd.read_sql("SELECT * FROM city_hour", conn)
```

In [16]:
```python
df2
```

Out[16]:

	ï»¿City	Datetime	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	O3	Benzene	Toluene	Xylene	AQI	AQI_
0	Ahmedabad	01-01-2015			1.00	40.01	36.37		1.00	122.07		0.0	0.0	0.0		
1	Ahmedabad	01-01-2015			0.02	27.75	19.73		0.02	85.90		0.0	0.0	0.0		
2	Ahmedabad	01-01-2015			0.08	19.32	11.08		0.08	52.83		0.0	0.0	0.0		
3	Ahmedabad	01-01-2015			0.30	16.45	9.20		0.30	39.53	153.58	0.0	0.0	0.0		
4	Ahmedabad	01-01-2015			0.12	14.90	7.85		0.12	32.63		0.0	0.0	0.0		
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
59461	None	None	None	None	NaN	NaN	NaN	None	NaN	NaN	None	NaN	NaN	NaN	None	
59462	None	None	None	None	NaN	NaN	NaN	None	NaN	NaN	None	NaN	NaN	NaN	None	
59463	None	None	None	None	NaN	NaN	NaN	None	NaN	NaN	None	NaN	NaN	NaN	None	
59464	None	None	None	None	NaN	NaN	NaN	None	NaN	NaN	None	NaN	NaN	NaN	None	
59465	None	None	None	None	NaN	NaN	NaN	None	NaN	NaN	None	NaN	NaN	NaN	None	

59466 rows × 16 columns

In [17]:
```python
df2.head(10)
```

Out[17]:

	ï»¿City	Datetime	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	O3	Benzene	Toluene	Xylene	AQI	AQI_Bucke
0	Ahmedabad	01-01-2015			1.00	40.01	36.37		1.00	122.07		0.0	0.0	0.0		
1	Ahmedabad	01-01-2015			0.02	27.75	19.73		0.02	85.90		0.0	0.0	0.0		
2	Ahmedabad	01-01-2015			0.08	19.32	11.08		0.08	52.83		0.0	0.0	0.0		
3	Ahmedabad	01-01-2015			0.30	16.45	9.20		0.30	39.53	153.58	0.0	0.0	0.0		
4	Ahmedabad	01-01-2015			0.12	14.90	7.85		0.12	32.63		0.0	0.0	0.0		
5	Ahmedabad	01-01-2015			0.33	15.95	10.82		0.33	29.87	64.25	0.0	0.0	0.0		
6	Ahmedabad	01-01-2015			0.45	15.94	12.47		0.45	27.41	191.96	0.0	0.0	0.0		
7	Ahmedabad	01-01-2015			1.03	16.66	16.48		1.03	20.92	177.21	0.0	0.0	0.0		
8	Ahmedabad	01-01-2015			1.47	16.25	18.02		1.47	16.45	122.08	0.0	0.0	0.0		
9	Ahmedabad	01-01-2015			2.05	13.78	16.08		2.05	15.14		0.0	0.0	0.0		

In [19]:
```python
df2.head(10000).to_csv('df2.csv')
```

In [ ]: